

SHORT BIO. My research focuses on Causal Decision Making and Causal Effect Identification. I study how to graphically model a wide range of real-world problems in a causal manner and make optimal decisions. In particular, my research explores methods for making robust decisions, even when modeling with incomplete information, and decision-making processes in sequential settings.

EDUCATIONS

Graduate School of Data Science, Seoul National University

Integrated M.S. and Ph.D. in Data Science

Advisor: Prof. [Sanghack Lee](#).

Seoul, Republic of Korea
 Sep 2022 - Aug 2027 (expected)

Department of Mathematics and Statistics, Sejong University

B.S. in Applied statistics and Mathematics (Double major), *Cum Laude*

Advisors: Prof. [Ji Eun Lee](#) and [Jin Hee Yoon](#).

Seoul, Republic of Korea
 Mar 2016 - Aug 2022

PUBLICATIONS

* for joint first authorship and [†] for corresponding author.

[Min Woo Park](#) and [Sanghack Lee](#)[†], *Counterfactual Structural Causal Bandits*, International Conference on Learning Representations (ICLR), 2026.

[Min Woo Park](#), Andy Ardit, Elias Bareinboim[†], [Sanghack Lee](#)[†], *Structural Causal Bandits under Markov Equivalence*, Neural Information Processing Systems (NeurIPS), 2025., Columbia CausalAI Laboratory, Technical Report (R-122)

[Min Woo Park](#) and [Sanghack Lee](#)[†], *On Transportability for Structural Causal Bandits*, arXiv preprint, 2025.

Yonghan Jung[†], [Min Woo Park](#), and [Sanghack Lee](#)[†], *Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference*, Neural Information Processing Systems (NeurIPS), 2024.

[Min Woo Park](#)^{*}, and [Ji Eun Lee](#)^{*†}, *Computation of the iterated Aluthge, Duggal, and Mean transforms*, Filomat, 2023.

[Ji Eun Lee](#)^{*†} and [Min Woo Park](#)^{*}, *Convergence of the iterated mean transforms of a 2×2 matrix*, under review.

AWARDS AND HONORS

BK21 Scholarship	2024, 2025
Fourth place award The 9th College of Natural Sciences Academic Conference, Sejong University ▷ Mitigating spurious regression in time-series data via Granger causality.	Nov 2021
First place merit scholarship Department of Mathematics and Statistics, Sejong University	Fall 2021
Third place award The 8th College of Natural Sciences Academic Conference, Sejong University ▷ The significance of the Euler's number and its applications.	Nov 2020
First place merit scholarship Department of Mathematics and Statistics, Sejong University	Fall 2020

PROJECTS

Towards More Scalable Score-based Causal Discovery LG AI Research ▷ Working on developing scalable score-based causal discovery algorithms.	Sep 2025 - Aug 2026
Hyundai NGV Data Boot-Campus Hyundai Motors ▷ Development of high-voltage battery performance prediction technology.	July 2025 - Sep 2025

- ▷ Development of an intelligent system for predicting vulnerable regions of automotive body panel stiffness through differential geometry-based automatic curvature analysis.

TEACHING ASSISTENTS

Causal Inference for Data Science	Spring, Fall 2024, Spring 2025, Spring 2026
▷ Bayesian Networks and Pearl's Graphical Causal Inference Framework.	
Machine Learning and Deep Learning for Data Science II	Fall 2023
▷ Kernel Methods, Gaussian Processes, and Related Topics for Machine Learning	
Basic Mathematics for Artificial Intelligence	Spring 2022
▷ Linear Algebra, Basic Statistics, and Probability Theory for Artificial Intelligence.	

PROFESSIONAL TALKS AND POSTERS

Counterfactual Structural Causal Bandits

ICLR 2026 poster session Rio de Janeiro, Brazil	Apr 2026
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Structural Causal Bandits under Markov Equivalence

ExploreCSR poster session Seoul, Republic of Korea	Dec 2025
NeurIPS 2025 poster session San Diego, USA	Dec 2025
JKAIA 2025, NeurIPS preview session Seoul, Republic of Korea	Nov 2025
SNU GSDS student research seminar Seoul, Republic of Korea	Apr 2025

Complete Graphical Criterion for Sequential Covariate Adjustment in Causal Inference

NeurIPS 2024 poster session Vancouver, Canada	Dec 2024
SNU GSDS student research seminar Seoul, Republic of Korea	Dec 2024

ACADEMIC SERVICES

Conference Reviewer: 2026 ICLR, ICML, 2025 ICML

REFERENCE

Sanghack Lee Seoul National University	sanghack@snu.ac.kr
Graduate School of Data Science	
Associate professor	
Ji Eun Lee Sejong University	jieunlee7@sejong.ac.kr
Department of Mathematics and Statistics	
Associate Professor	
Jin Hee Yoon Sejong University	jin9135@sejong.ac.kr
Department of Artificial Intelligence and Information Technology	
Associate Professor	

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